

Model Training_Conversion

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1、Model Training

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References

After completing the dataset annotation tutorial content, we can use the board to start training the model.

This tutorial only introduces CLI model training and conversion, you can refer to the official website for Python case modifications.

1、Model Training

Use CLI commands to directly train the model

```
cd /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data
```

```
yolo detect train data=orange.yaml model=yolo26n.pt epochs=100 imgsz=640
```

`data`: Dataset configuration file

`model`: Pre-trained model file

`epochs`: Number of training rounds

`imgsz`: Specify input image size

```
jetson@ubuntu:~/ultralytics/ultralytics/data/yahboom_data/orange_data$ yolo detect train data=orange.yaml model=yolo26n.pt epochs=100 imgsz=640
Ultralytics 8.4.21 Python-3.12.3 torch-2.10.0 CUDA:0 (NVIDIA Thor, 125772MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None,
0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=orange.yaml, deepsort=0.0, deterministic=True,
ut=0.0, dynamic=False, embed=None, end2end=None, epochs=100, ema=False, imgsz=[640, 640], iou=0.7, keras=False, lr=0.01, lr_scheduler=cos, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=train3, nbs=
ose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=False, save_dir=runs/detect/train3, save_frames=False, save_json=False, save_period=10,
e, simplify=True, single_cls=False, source=None, split=val, static=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.1, warmup_nbs=16, warmup_t=5,
Overriding model.yaml nc=80 with nc=1

      from  n  params module
      -1  1    464   ultralytics.nn.modules.conv.C2f
      1  -1  1   4672   ultralytics.nn.modules.conv.C2f
      2  -1  1   6640   ultralytics.nn.modules.conv.C2f
      3  -1  1  36992   ultralytics.nn.modules.conv.C2f
      4  -1  1  26080   ultralytics.nn.modules.conv.C2f
      5  -1  1 147712   ultralytics.nn.modules.conv.C2f
      6  -1  1   87040   ultralytics.nn.modules.conv.C2f
      7  -1  1 295424   ultralytics.nn.modules.conv.C2f
      8  -1  1 346112   ultralytics.nn.modules.conv.C2f
      9  -1  1 164608   ultralytics.nn.modules.conv.C2f
     10  -1  1 249728   ultralytics.nn.modules.conv.C2f
     11  -1  1      0   torch.nn.modules.upsampling.Upsample
     12  [-1, 6] 1      0   ultralytics.nn.modules.conv.Concat
     13  -1  1 119808   ultralytics.nn.modules.conv.C3k2
```

```

YOLO26n summary (fused): 122 layers, 2,375,031 parameters, 0 gradients, 5.2 GFLOPs
AP50-95): 100% 

| Class | Images | Instances | Box(P | R  | mAP50 |
|-------|--------|-----------|-------|----|-------|
| all   | 1/1    | 5.3it/s   | 0.2s  | 20 | 20    |
|       |        |           | 0.997 | 1  | 0.995 |


0.957
Speed: 0.1ms preprocess, 2.9ms inference, 0.0ms loss, 0.3ms postprocess per image
Results saved to /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train
Learn more at https://docs.ultralytics.com/modes/train
jetson@ubuntu:~/ultralytics/ultralytics/data/yahboom_data/orange_data$

```

Finally, the trained model-related files are stored in the
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train folder.

Note: Each time training is run, the number after train will automatically increment by 1, please record this path, and use this new model path later.

2、Model Conversion

The final model will be generated in the runs folder: generally select the best.pt file for use

```
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs
```

Convert PyTorch format model to TensorRT:

```

cd
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
ls

```

```
yolo export model=best.pt format=engine
```

```

jetson@ubuntu:~/ultralytics/ultralytics$ cd /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
jetson@ubuntu:~/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights$ ls
best.pt  last.pt
jetson@ubuntu:~/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights$ yolo export model=best.pt format=engine
2026-03-27 11:34:56.842414661 [W:onnxruntime:Default, device_discovery.cc:164 DiscoverDevicesForPlatform] GPU device discovery failed: device_discovery.cc:89 ReadFileContents Failed to open file: "/sys/class/drm/card1/device/vendor"
WARNING ⚠️ TensorRT requires GPU export, automatically assigning device=0
Ultralytics 8.4.21 Python-3.10.12 torch-2.5.0a0+872d972e41.nv24.08 CUDA:0 (Orin, 62841MiB)
YOLO26n summary (fused): 122 layers, 2,375,031 parameters, 0 gradients, 5.2 GFLOPs

```

Wait for the model conversion to complete, best.engine and best.onnx files will be generated in the current directory.

```
ls
```

```
Export complete (110.3s)
Results saved to /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
Predict:      yolo predict task=detect model=best.engine imgsiz=640
Validate:     yolo val task=detect model=best.engine imgsiz=640 data=orange.yaml
Visualize:    https://netron.app
💡 Learn more at https://docs.ultralytics.com/modes/export
jetson@ubuntu:~/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights$ ls
best.engine  best.onnx  best.pt    last.pt
jetson@ubuntu:~/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights$
```

References

<https://docs.ultralytics.com/modes/train/>